

Predicting Flight Delays Using Machine Learning

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Abstract

Flight delays remain a persistent challenge for the aviation industry, generating high costs and disrupting operations across interconnected networks. Existing monitoring and scheduling tools provide valuable oversight but often lack predictive accuracy and cross-stakeholder coordination, limiting their effectiveness in disruption management. This study develops and evaluates an AI-enabled machine learning framework that integrates operational and meteorological data to forecast delays more reliably. Using U.S. domestic flight records and NOAA weather data for 2024, including a case study at Louisville Muhammad Ali International Airport, we apply classification and regression models to predict on-time performance and delay minutes. Ensemble methods, particularly Random Forest with SMOTE balancing, achieve superior results, detecting delayed flights with 94.7% accuracy and reducing mean absolute error in regression tasks to 4.79 minutes. Beyond technical gains, the framework demonstrates how AI-driven prediction can enhance collaborative decision-making by enabling shared situational awareness across airlines, airports, and air traffic control, strengthening resilience and efficiency in aviation operations.

1. Introduction

We all use airlines today whether we are going to a meeting in a distant city, going on a trip or vacation or attending a conference or travelling international to board a cruise. Pilots and flight attendants use the airlines to fly from their present location to the airport where they are starting the trip. Airline delays cause problems for travelers and cost the airlines billions of dollars annually (Air Travel Consumer Report – ATCR, 2024). This paper performs data analysis and data exploration on US domestic flight delays, as tracked by the Bureau of Transportation Statistics (BTS) for 2024. A computer simulation study (SDF Study) contains flights that depart from SDF

(Louisville Muhammad Ali International Airport) and return to SDF. The SDF study only looks at airports where the number of flights is greater than 300 (origin, destination pairs) outgoing from SDF and incoming to SDF. Some reports and graphs are included to help one understand the flight data for both flight data sets. The SDF study merges the flight data with NOAA weather data by airport, flight date, and hour. Then the weather data is added for each flight's outgoing and incoming airport. Last, several machine learning algorithms are processed to predict results for aircraft arrival delays and aircraft arrivals delay minutes. Cancelled and diverted flights are removed from the SDF study.

The topic explored in this study contributes to the ongoing discourse on the innovative, collaborative, and sustainable development of organizations through the application of AI and BD technologies. As the operational and economic challenges of the aviation industry intensify, the strategic use of machine learning for predictive analytics becomes increasingly relevant. This research addresses the problem of flight delays by developing robust predictive models based on a large-scale integration of transportation and weather data. The approach demonstrates how complex, real-world datasets can be transformed into actionable intelligence, supporting informed decision-making within time-sensitive and resource-constrained environments.

By employing advanced data engineering techniques, including feature-rich integration of meteorological conditions and operational indicators, as well as classification enhanced by SMOTE and regression algorithms, the study provides a scalable and transferable framework. The combination of methodological rigor and practical insight offers a compelling example of how AI and BD can be mobilized to reduce inefficiencies, support sustainable resource allocation, and enable more resilient organizational responses. Furthermore, the research process itself reflects a collaborative paradigm - blending interdisciplinary expertise and diverse data

ecosystems to generate innovative outcomes with tangible societal and organizational impact.

The paper is organized as follows. Section 2 provides motivation for the study. Section 3 reviews the relevant literature. Flight definitions are explained in Section 4. Computer simulations, including the methods and results, are discussed in Section 5. Finally, Section 6 concludes the paper.

2. Motivation

Airline operations depend on the constant coordination of multiple actors - airlines, airports, and air traffic control - who must jointly manage complex schedules and scarce resources. Despite significant investments in IT systems, current approaches often operate in silos, focusing on local efficiency rather than network-wide resilience. Industry initiatives such as Airport Collaborative Decision Making (A-CDM) have improved information sharing, but they remain largely descriptive and reactive, lacking predictive and data-driven capabilities (Eurocontrol, 2016; ICAO - Annual Report 2018). This fragmentation limits the industry's ability to anticipate and mitigate cascading disruptions. Flight delays, therefore, remain not only costly but also symptomatic of insufficient collaboration across the aviation ecosystem.

Although machine learning has recently been applied to flight delay prediction, most studies focus on algorithmic performance and do not explicitly link predictive analytics to collaborative decision-making frameworks (Ribeiro et al., 2023; Al-Sharif & Naveh, 2024).

Our study addresses this gap by developing predictive models that integrate large-scale transportation and weather data, enabling stakeholders to anticipate disruptions more effectively. By providing accurate forecasts and actionable insights, the proposed framework supports collaborative decision-making in time-sensitive environments. In this way, the study directly contributes to the broader theme of collaboration systems and technologies, showing how AI-enabled prediction can enhance

Table 1 presents the number, percentage, and delay minutes for U.S. domestic flights and for flights to and from SDF, broken down by on-time performance and categories of delay. It is evident that Air Carrier Delay (6.54%), NAS Delay (5.55%), and Late Aircraft Delay (7.58%) are the primary contributors to flight delays. Flights arriving at or departing from SDF follow a very

coordination, resource allocation, and organizational resilience across interconnected aviation stakeholders. In addition, our study demonstrates superior results compared with those discussed in the literature review and includes statistical tests to assess whether the differences between methods are significant. However, numeric results across studies are not directly comparable, given differences in datasets (e.g., time periods, local airports, airlines), feature selections, preprocessing steps, and evaluation protocols, highlighting the importance of context-specific evaluation.

Figures 1 and 2 illustrate key geographical aspects of the study. Figure 1 displays the map of U.S. domestic flights, excluding Hawaii, Alaska, and Puerto Rico. Figure 2 shows the locations of the 23 airports included in the analysis.

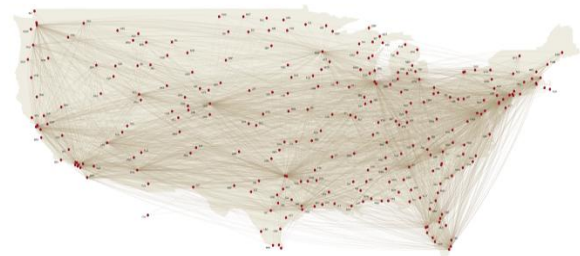


Figure 1. US Domestic Flights Map.

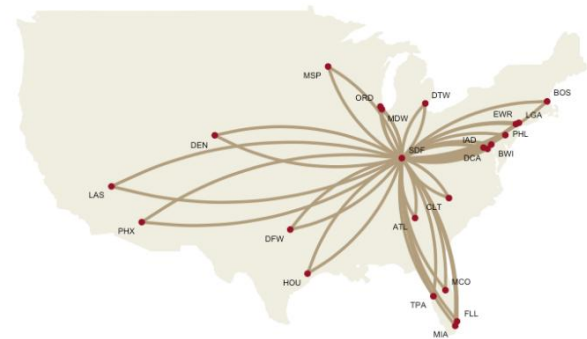


Figure 2. SDF Study Map.

similar pattern. Surprisingly, weather and security-related delays contribute relatively little to overall delays. In 2024, SDF handled 48,006 flights, of which 484 were cancelled and 84 diverted. After excluding cancelled and diverted flights, 47,436 flights remained for analysis in the SDF study.

Table 1. US Domestic and SDF Flight Reports – On Time Performance and Delays.

Category	US Domestic Reports			SDF Reports		
	Operations	%	Delay Minutes	Operations	%	Delay Minutes
On Time	5,515,281	77.91	NA	38,299	79.78	NA
Air Carrier Delay	463,198	6.54	35,823,263	3,038	6.33	247,287
Weather Delay	53,628	0.76	6,195,873	343	0.71	31,390
National Aviation System (NAS) Delay	392,835	5.55	19,621,964	2,597	5.41	122,697
Security Delay	3,920	0.06	179,928	31	0.06	1,481
Late Aircraft Delay	536,384	7.58	41,972,684	3,128	6.52	246,175
Cancelled	96,315	1.36	NA	486	NA	NA
Diverted	17,500	0.25	NA	84	NA	NA
Totals	7,079,061	100.00	103,794,712	48,006	100.00	649,030

3. Literature Review

Flight delays are a pervasive issue within the aviation industry, posing significant operational and economic challenges. Recent research reflects a growing focus on predictive modeling using machine learning techniques to proactively address delay phenomena.

3.1. Methods and Attributes Used

One significant contribution to this field is the work by Kandula et al. (2024), who proposed a Neural Fusion Network (NFN) that integrates Long Short-Term Memory (LSTM) and Recurrent Neural Network (RNN) architectures. Their model leverages a rich dataset that includes historical flight logs, weather variables (e.g., wind speed, visibility, precipitation), flight schedule details, airport congestion metrics, and aircraft-specific parameters. The integration of both long-term and short-term temporal dependencies enabled the NFN model to achieve a 91% prediction accuracy, outperforming standalone models. Yi et al. (2021) developed a stacking-based ensemble learning model that synthesized predictions from base learners such as KNN, Random Forest, and Logistic Regression. Their feature space included categorical variables such as airline code, origin and destination airport, and day of the week, as well as numerical variables like scheduled departure time, delay durations, and distance. Feature selection was conducted using the Boruta algorithm to remove redundant predictors, while SMOTE was employed to balance class distributions in delay outcomes.

In the study by Yazdi et al. (2020), the authors implemented an LSTM network trained with the Levenberg–Marquardt algorithm, which is well-suited for complex regression tasks. The model used variables such as weather conditions, scheduled and actual departure/arrival times, and delay types. Temporal

dependencies in features like arrival delay, departure delay, and historical delay frequencies were explicitly modeled. The inclusion of time series inputs enabled the model to capture seasonal and daily patterns in flight operations. Sahu et al. (2023) used a Support Vector Machine (SVM) framework trained on a compact yet diverse set of predictors in another study. These included weather conditions, airline identifiers, flight route distance, scheduled time blocks, and previous delay history. The authors emphasized the benefit of using structured data from government transportation records and demonstrated that even with fewer variables, meaningful patterns could be captured for delay classification.

Across all studies, a common theme is the strategic selection and engineering of variables. Variables such as departure time, day of the week, aircraft type, air traffic volume, weather status, and flight leg (origin-destination pair) were repeatedly found to influence delay outcomes. Some studies also introduced engineered features like time-of-day slots (morning, afternoon, evening), weekend flags, or lagged delay values, which helped boost model generalization. Visualization techniques such as scatter plots, interaction graphs, and heatmaps provided additional clarity on the relationship between variables and delays. For example, plotting scheduled vs. actual departure times highlighted systemic bottlenecks at peak hours, while heatmaps illustrated which airlines or routes experienced higher delays across different weekdays.

Taken together, these works show that variable selection plays a critical role in predictive performance. Models that incorporate a comprehensive, well-curated feature set; especially those that combine operational, meteorological, and temporal variables; are more capable of producing accurate and actionable delay forecasts. As predictive frameworks become more mature, future work may explore real-time variable ingestion, transfer learning across airports, and

integration of external datasets such as live weather streams or airline maintenance logs.

Flight delays remain one of the most significant challenges facing the aviation industry. They throw off flight schedules, frustrate passengers, and cost airlines a lot of money. As air traffic increases, researchers have been working to better understand, predict, and reduce delays by examining many factors, such as airline scheduling, airport operations, weather, and technical issues. Some of the earliest studies focused on how airlines and airports are run. For example, Brueckner et al. (2021) looked at adding extra time into flight schedules, which can help prevent a delay from affecting later flights. However, adding too much time also makes the system less efficient. In another study, Calzada and Fageda (2023) found that airports used mainly by a few large airlines tend to have more delays, especially when the airspace is crowded.

Weather is still one of the most common reasons flights get delayed. Borsky and Unterberger (2019) broke weather into sudden events (like storms) and slower ones (like fog) and found that both types can lead to severe delays. Skultety et al. (2021) extended this idea by studying how climate change leads to more extreme weather in Europe, particularly storms, making managing flights even harder. Milosevic et al. (2024) added a seasonal perspective and used artificial neural networks to improve predictions. Predictive models have become more popular as more flight and operations data becomes available. Henriques and Feiteira (2018) used decision trees and neural networks to model delays at Atlanta's primary airport with promising results. Thiruthuvanathan et al. (2025) later introduced a deep learning model that used LSTM networks to detect patterns in delayed data over time, which helped make better predictions.

Bisandu and Moulitsas (2024) took a creative approach, using Deep Operator Networks to predict delays. They improved their model using a nature-inspired optimization algorithm called the Gradient Mayfly Optimization Algorithm. This helped adjust the model's weights and reduced prediction errors, which shows that combining optimization tools with deep learning can lead to more accurate results for difficult problems like flight delays. Recent work has also focused on making models easier to understand. Pineda-Jaramillo et al., 2024 combined data types, such as weather, location, and airport operations, into machine-learning models that explain their results. Using tools like SHAP and Sobol, they showed which factors had the most significant impact on delays, helping decision-makers for more effective planning. Other researchers have taken a probabilistic approach. Beltman et al. (2025) trained models like Random Forest and CatBoost to predict whether a flight would be delayed and how likely and long the delay might be.

They reported that at the 90-minute horizon, MAE and RMSE were 8.46 and 11.91 minutes, respectively. This forecasting is especially helpful at busy hub airports with tight timing. Spatiotemporal models have added another layer of insight. Mamdouh et al. (2023) tested several machine learning methods and found that FDPP-ML stacking regression achieved the best prediction accuracy, with an MAE of 16.69 minutes.

Zhang et al. (2021) analyzed flight delay patterns across more than 200 airports in China using tensor factorization. Their study revealed clusters of airports that shared similar delay behaviors, which could help make smarter choices about staffing levels and where to invest in airport improvements. It is still essential to examine what is causing flight delays. Mokhtarimousavi and Mehrabi (2023) used statistical and machine-learning models to uncover less obvious factors behind delays. Their research showed that things like flight schedules, geographical specification of the origin airport, and even the time of day can all play a role, not just weather or air traffic. Passenger behavior has also been explored. Lee and Yip (2017) looked at how delays influence which airline travelers choose. While delays mattered, passengers often picked flights based on timing and service quality, especially when comparing budget and full-service airlines. Some studies have shifted from analyzing delays to preventing them. Jayaraman et al. (2022) developed a flight control system that can adjust if a part fails, helping avoid delays during landing. Lin et al. (2021) worked on time-delay compensation in-flight systems, which is critical for safety in extreme flight conditions. Overall, research has moved from simply identifying delay causes to predicting them more accurately and even preventing them through better systems. There is now a strong push toward making models that are precise, easy to understand, and flexible enough to use in real-world airport and airline operations.

3.2. Carrier Delays Cost Studies and Other Major Delays

Although the study for 2023 reports that the cost of U.S. domestic flight delays decreased by 0.4% compared to the previous year, delays still cost airlines several billion dollars in additional expenses (Airlines for America, 2024). For example, the average cost of aircraft block time (taxi plus airborne) for U.S. passenger airlines was \$100.80 per minute. Fuel expense, the largest cost component, declined by 11.6% to \$37.27 per minute. Labor costs, the second-largest component, increased by 12.7% to \$32.68 per minute, while maintenance expenses rose by 7.2% to \$16.88 per minute.

Using an estimated value of \$47 per hour for a passenger's time (DOT FAA, 2015), flight delays are also believed to have cost air travelers billions of dollars. FAA/Nextor (2024) estimated that the total annual cost of delays—including direct costs to airlines and passengers, lost demand, and indirect costs—reached \$33 billion in 2019.

A study on the 2022 flight disruptions estimated the economic impact at \$30–34 billion, with approximately 200 million passengers experiencing delays or cancellations. Of these costs, 37% were borne by passengers and 32% by the airlines (Junginger, 2023). Guy (2010) previously estimated that domestic flight delays cost the U.S. economy \$32.9 billion annually.

In response to persistent service issues, the Department of Transportation fined JetBlue \$2 million for chronic delays (Chung, 2025). A chronic delay is defined as a flight that operates at least 10 times per month and arrives more than 30 minutes late more than 50% of the time. Intarasuwan (2025) reports that major delays and cancellations have significantly affected travelers at Newark Liberty International Airport in New Jersey. A chain of delays came after a number of system outages, including when controllers' screens essentially went dark for up to 90 seconds — causing them to lose the ability to track aircraft at a key facility handling traffic in and out of Newark. At one point, arriving flights were delayed by up to six hours, while departures faced delays of nearly four hours. In addition to the long departure delays these statements demand a call for changes in the tracking systems at Newark airport and are of grave concern for safety in our airline tracking and communication systems.

In parallel with advances in predictive modeling, the aviation industry has emphasized the role of Collaborative Decision Making (CDM) to improve operational efficiency. CDM - promoted by ICAO, FAA, IATA, Eurocontrol, and other groups - highlights the importance of shared information, transparency, and joint decision-making among stakeholders such as airlines, airports, and air traffic management authorities. While existing CDM practices have relied primarily on operational reporting and manual coordination, they often lack the predictive, data-driven dimension that can anticipate disruptions before they propagate across the network (Jeeradist., 2023; Ding et al., 2024). Other studies highlight AI's potential as an "information broker" to enhance shared situational awareness across stakeholders (Geske, et al., 2025). Our study builds on and contributes to this emerging stream of research by showing how machine learning models can operationalize and enhance CDM: providing stakeholders with real-time forecasts of delays, supporting shared situational awareness, and

enabling more coordinated and collaborative responses to system disruptions.

4. Flight Definitions

4.1. Flight Delay

A flight is counted as "on-time" if it operated less than 15 minutes after the scheduled time shown in the carriers' Computerized Reservations Systems (CRS) (BTS, 2021). A delayed flight is a flight that arrives 15 minutes or greater based on the scheduled arrival time of the flight. When multiple causes are assigned to one delayed flight, each cause is prorated based on delayed minutes it is responsible for. The displayed numbers are rounded and may not add up to the total. (DOT BTS, 2019).

4.2. Causes of Delay (BTS, 2021)

- Air Carrier: The cause of the cancellation or delay was due to circumstances within the airline's control (e.g. maintenance or crew problems, aircraft cleaning, baggage loading, fueling, etc.).
- Extreme Weather: Significant meteorological conditions (actual or forecasted) that, in the judgment of the carrier, delays or prevents the operation of a flight such as tornado, blizzard or hurricane.
- National Aviation (Airspace) System (NAS) (FAA, 2023): Delays and cancellations attributable to the national aviation system that refer to a broad set of conditions, such as non-extreme weather conditions, airport operations, heavy traffic volume, and air traffic control.
- Late-arriving aircraft: A previous flight with same aircraft arrived late, causing the present flight to depart late.
- Security: Delays or cancellations caused by evacuation of a terminal or concourse, re-boarding of aircraft because of security breach, inoperative screening equipment and/or long lines in excess of 29 minutes at screening areas.

4.3. Cancelled and Diverted Flight (Office of Aviation Consumer Protection, 2024)

- A "cancelled" flight is a flight that was not operated but was in the carrier's computer reservation system within 7 days of the scheduled departure.
- A "diverted" flight is a flight which is operated from the scheduled origin point to a point other than the scheduled destination point in the carrier's published schedule.

5. Computer Simulation of Flight Delays: Methods and Results

5.1. Data Sources and Attributes Used

Two real-world data sources were used in this paper (BTS airline flight and arrival performance data and NOAA weather data). The first BTS data source can be downloaded from this link (https://www.transtats.bts.gov/DL_SelectFields.aspx?gnoyr_VQ=FGJ&QO_ful46_anzr=b0-gvzr). Two sets of flight data were collected. The first flight dataset contains all US domestic flights in 2024 that were flown, canceled or diverted as shown in Table 1. Figure 1 shows US domestic flights (a) without its territories for the year 2024, (b) flights within Longitude (-130, -60) degrees and Latitude (25, 50) degrees, and (c) airports having > 300 flights (origin, destination pairs) for 2024 – about 1 per day. The US map showing flights was filtered to help the reader visualize the airline traffic between the major US domestic hubs. The major hubs include Atlanta (ATL), Dallas (DFW), Chicago (ORD), Denver (DEN), Charlotte (CLT), Los Angeles (LAX), Phoenix (PHX), Las Vegas (LAS), Seattle (SEA), New York (LGA), Orlando (MCO), Boston (BOS), and Washington (DCA). The numeric and nominal attributes represented Month, Day of Month, Day of Week,

Flight Date, Origin Airport ID, Destination Airport ID, Scheduled Departure Time, and Scheduled Arrival Time to name a few. The second flight dataset contains flights in and out of the Louisville SDF airport in 2024, where the flight pairs (in, out) are > 300 flights for each airport per year.

The second data source contains hourly weather NOAA weather data for each airport by year and Station ID (airport) and it can be downloaded from this NOAA link (<https://www.ncei.noaa.gov/data/local-climatological-data/access/>). It includes variables such as Latitude, Longitude, Elevation, Station ID, hourly measurements of Temperature, Precipitation, Humidity, Pressure, Sky Condition, Wind Speed, and Wind Gust Speed. For more details, please refer to the above links.

The computer simulation used 2024 flight data for flights to and from Louisville (SDF), involving 11 airlines (Table 2) and 23 airports (Table 3). Airports with fewer than 300 flights in 2024 were arbitrarily excluded from the analysis. The original dataset contained 109 variables and 47,436 flight records, including on-time, delayed, diverted, and cancelled flights. However, diverted and cancelled flights were removed from analysis, leaving only on time and delayed flights. This filtering resulted in 47 variables used for modeling, 46 independent variables and one dependent variable.

determine whether a flight was on time or delayed. A flight was considered on time, if it arrived within 15 minutes of the scheduled arrival. The second scenario used regression analysis to predict the number of minutes a flight was delayed.

5.2. Simulation Scenarios

The simulation included two scenarios. The first scenario focused on a binary classification task to

Table 2. Airlines Included.

1	AA – American Airlines	7	WN – Southwest Airlines
2	DL – Delta Airlines	8	YX – Republic Airlines
3	G4 – Allegiant Air	9	9E – Endeavor Air
4	MQ – Envoy Air	10	OO – SkyWest Airlines
5	NK – Spirit Airlines	11	UA – United Airlines
6	OH – PSA Airlines		

First Scenario: Classification – On-time vs. Delayed Flights

The dependent variable for this scenario, *ArrDel15*, coded flights as 0 (on time) or 1 (delayed). Of the 47,435 flights, 38,299 were on time and 9,136 were delayed, revealing a significant class imbalance. To address this imbalance, the SMOTE algorithm was applied, increasing the number of delayed flights to 36,544 while retaining the original number of on-time

Table 3. Airports Included.

1	SDF	7	ORD	13	BOS	19	BOS
2	DFW	8	MCO	14	PHX	20	DTW
3	MIA	9	LAS	15	TPA	21	EWB
4	CLT	10	BWI	16	DCA	22	IAD
5	ATL	11	DEN	17	PHL	23	MSP
6	FLL	12	HOU	18	LGA		

flights. This led to improved classification performance. Independent variables used in several simulation scenarios included *DepDel15* and *DepDelay*, representing departure delay indicator (set to 1 if the delay is 15 minutes or more, and 0 otherwise) and the difference in minutes between scheduled and actual departure time, respectively. Early departures were encoded as negative numbers. The underlying assumption was that if the arriving flight is late, the departing flight is likely to be delayed as well. Further, if that plane is the same plane that is scheduled to leave

on, the plane may also be late leaving the airport and later arriving at its destination.

Attributes such as *Taxi Out*, *Wheels Off*, and arrival data such as *Wheels On*, *Taxi In* were not used, as they are highly correlated with the dependent variable.

The classification models included: Logistic Regression (LR), used as a baseline to which other methods were compared; k -Nearest Neighbors with $k=3, 5$, and 8 (using normalized Euclidean distance was used for numeric variables and Hamming distance for nominal ones); Bagging (Bootstrap aggregating ensemble method); Decision Tree (DT) based on Quinlan's algorithm; RepTree (a fast decision tree learner using reduced-error pruning); and Random Forest (RF), which consistently achieved the best results.

Model performance was evaluated using several metrics: overall classification accuracy, accuracy for on-time and delayed flights, area under the ROC curve (AUROC), and the Kappa statistic. Techniques significantly better than the baseline ($p = 0.05$) are marked with a superscript b , and those significantly worse are marked with w . Neural networks and support vector machines (SVMs) were also tested but excluded from further analysis due to long runtimes and comparable performance. A 66/34% train/test split confirmed the robustness of the results, yielding outcomes similar to 10-fold cross-validation, which was used in this study because the flight dataset was not chronologically ordered. Several variable reduction techniques were also tested, but none significantly improved classification accuracy.

Without using variables *DepDel15* and *DepDelay*, LR achieved a moderate overall accuracy of 81.3%. RF performed best overall (84.3%) and had the highest AUROC and Kappa values. Delayed flight detection rates were generally low, between 10.6% and 32.3%,

across models without SMOTE (Table 4). SMOTE improved all models. RF performed best with an overall accuracy of 90.6%, a delayed flight detection rate of 83.8%, AUROC of 0.958, and a Kappa of 0.81 (Table 5). Including *DepDel15* and *DepDelay* improved baseline performance. RF achieved the highest accuracy (93.9%), with improved delayed flight detection (76.5%) and AUROC (0.949). (Table 6). RF again led in performance with an overall accuracy of 96.0%, delayed flight detection at 94.7%, AUROC of 0.988, and Kappa of 0.92 (Table 7). All models improved substantially with SMOTE and the inclusion of all 47 variables.

Second Scenario: Regression, Predicting Flight Delays in Minutes

The goal in this scenario was to predict the actual number of minutes a flight was delayed. The dependent variable, *Delay in Minutes*, was numeric. The models evaluated were: Additive Regression (AR), used as the baseline; neural networks (NN), with decay set to True; Bagging; Random Forest (RF); and RepTree. Model performance was assessed using five metrics: mean absolute error (MAE), root mean squared error (RMSE), relative absolute error (RAE), root relative squared error (RRSE), and the correlation coefficient (CC).

AR and RepTree outperformed other models across all five metrics (Table 8). RF, in contrast to its strong classification performance, performed significantly worse in the regression scenario, with notably higher RMSE and lower correlation. Smaller error values indicate better predictive accuracy. Additive Regression, Bagging, and RepTree clearly outperformed the other methods across all evaluation metrics yielding the lowest error rates in terms of MAE and RMSE.

Table 4. Classification rates for the dataset with 45 variables (*DepDel15* and *DepDelay* excluded) without SMOTE.

Methods/ Rates	LR	3NN	5NN	8NN	Bagging	DT	REP Tree	RF
Overall [%]	81.3	78.7 ^w	80.7 ^w	81.5	82.5 ^b	81.8	81.7	84.3 ^b
On-Time [%]	98.2	91.6 ^w	94.8 ^w	97.9	96.7 ^w	93.6 ^w	96.0 ^w	98.0
Delayed [%]	10.6	24.6 ^b	21.9 ^b	12.6 ^b	23.1 ^b	32.3 ^b	21.7 ^b	26.7 ^b
AUROC	0.718	0.658 ^w	0.678 ^w	0.697 ^w	0.751 ^b	0.677 ^w	0.699 ^w	0.806 ^b
Kappa	0.13	0.19 ^b	0.21 ^b	0.15 ^b	0.26 ^b	0.31 ^b	0.23 ^b	0.33 ^b

Table 5. Classification rates for the dataset with 45 variables (*DepDel15* and *DepDelay* excluded) SMOTE.

Methods/ Rates	LR	3NN	5NN	8NN	Bagging	DT	REP Tree	RF
Overall [%]	79.9	87.0 ^b	87.8 ^b	87.8 ^b	88.6 ^b	88.0 ^b	87.6 ^b	90.6 ^b
On-Time [%]	87.3	88.7 ^b	89.7 ^b	91.6 ^b	96.1 ^b	92.5 ^b	95.1 ^b	97.0 ^b
Delayed [%]	72.0	85.1 ^b	85.8 ^b	83.9 ^b	80.8 ^b	83.1 ^b	79.8 ^b	83.8 ^b
AUROC	0.872	0.919 ^b	0.930 ^b	0.936 ^b	0.936 ^b	0.907 ^b	0.916 ^b	0.958 ^b
Kappa	0.60	0.74 ^b	0.76 ^b	0.76 ^b	0.77 ^b	0.76 ^b	0.75 ^b	0.81 ^b

Table 6. Classification rates for the dataset with 47 variables (including *DepDel15* and *DepDelay*) without SMOTE.

Methods/ Rates	LR	3NN	5NN	8NN	Bagging	DT	REP Tree	RF
Overall [%]	93.3	89.5 ^w	90.5 ^w	90.0 ^w	93.3	93.1	93.0	93.9 ^b
On-Time [%]	98.0	96.2 ^w	97.1 ^w	97.7 ^w	98.3 ^b	98.0	98.0	98.0
Delayed [%]	73.4	61.2 ^w	63.0 ^w	57.8 ^w	72.2 ^w	72.4	72.1 ^w	76.5 ^b
AUROC	0.937	0.870 ^w	0.889 ^w	0.903 ^w	0.936	0.880 ^w	0.913 ^w	0.949 ^b
Kappa	0.77	0.63 ^w	0.66 ^w	0.63 ^w	0.76	0.76	0.76	0.79 ^b

Table 7. Classification rates [for the dataset with 47 variables (including *DepDel15* and *DepDelay*) with SMOTE.

Methods/ Rates	LR	3NN	5NN	8NN	Bagging	DT	REP Tree	RF
Overall [%]	93.7	94.0 ^b	94.4 ^b	94.4 ^b	95.3 ^b	94.6 ^b	95.1 ^b	96.0 ^b
On-Time [%]	96.3	94.9 ^w	95.2 ^w	95.6 ^w	97.4 ^b	96.9 ^b	97.2 ^b	97.2 ^b
Delayed [%]	91.0	93.1 ^b	93.7 ^b	93.1 ^b	92.1 ^b	92.2 ^b	92.8 ^b	94.7 ^b
AUROC	0.978	0.967 ^w	0.973 ^w	0.976 ^w	0.982 ^b	0.958 ^w	0.975 ^w	0.988 ^b
Kappa	0.87	0.88 ^b	0.89 ^b	0.89 ^b	0.91 ^b	0.89 ^b	0.90 ^b	0.92 ^b

Table 8. Prediction error rates for the dataset with 47 variables (including *DepDel15* and *DepDelay*).

Methods/ Errors	AR	NN	Bagging	RF	Rep Tree
MAE	4.93	5.45 ^w	4.79	8.37 ^w	4.97
RMSE	10.72	11.75	11.58	30.16 ^w	11.48
RAE	22.72	25.06 ^w	22.05 ^b	38.51 ^w	22.91
RRSE	19.87	21.46 ^w	20.95	52.80 ^w	21.11
Correlation	0.98	0.98	0.98	0.90 ^w	0.98

6. Conclusion

This study develops and evaluates a scalable, simulation-based framework for forecasting flight delays using real-world aviation data from 2024 for flights departing from and arriving at the SDF airport. By applying a suite of machine learning algorithms—including *k*-nearest neighbors, decision trees, ensemble methods (Random Forest and Gradient Boosting), and techniques to address class imbalance such as SMOTE—we demonstrate the power of data-driven approaches for predictive modeling in transportation contexts.

Among the tested models, ensemble methods outperformed others. RF and Gradient Boosting

achieved the highest classification performance across all measures such as overall, on time, delayed classification accuracy rates, as well as AUROC, and Kappa statistic. The application of SMOTE significantly improved the models' ability to handle imbalanced datasets, enhancing their generalizability and robustness. In regression analyzes, AR, Bagging, and RepTree also delivered the lowest MAE (4.93, 4.79, and 4.97 minutes), RMSE (10.72, 11.58, and 11.48 minutes), and highest correlations (0.98, 0.98, 0.98), underscoring their effectiveness in capturing complex, nonlinear patterns in flight delay dynamics. Our study demonstrates that the proposed predictive framework achieves superior accuracy compared with several previous approaches, providing actionable

insights for aviation stakeholders. While direct comparison with other studies is limited due to differences in datasets, features, and evaluation methods, the results highlight the effectiveness of our context-specific, data-driven approach in anticipating flight disruptions. Beyond technical performance, the framework offers tangible, practical value for airline operations and transportation planning. Accurate delay forecasting supports better gate allocation, crew scheduling, customer communication, and overall resource optimization. These benefits contribute to both organizational efficiency and improved passenger experience.

In a broader sense, the study exemplifies how the integration of big data analytics and AI methods can lead to sustained innovation and collaborative development in data-intensive industries. By translating predictive insights into operational improvements, organizations can move toward more resilient, adaptive, and data-informed decision-making cultures. This work thus highlights not only methodological advancements, but also their strategic potential to support long-term, cross-organizational transformation.

Importantly, the framework should be viewed not merely as a technical solution, but as a collaboration enabler. By consolidating diverse data streams into a unified predictive model, the system provides stakeholders with a common reference point and shared situational awareness. Such collaborative intelligence reduces information asymmetries, aligns operational strategies, and supports joint decision-making under uncertainty. Compared with conventional scheduling tools and stand-alone predictive models, our framework contributes by bridging predictive analytics with collaborative decision-making. This dual contribution - technical accuracy and collaborative applicability - makes the approach superior to existing industry and research solutions. Looking ahead, the integration of AI-driven prediction into collaborative systems offers a promising pathway not only for aviation but also for other high-stakes domains - such as logistics, healthcare, and emergency management - where resilience depends on timely, coordinated decisions.

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